

Conditional Effective GR Completion of UBT

Ing. David Jaroš

26 July 2026

Abstract

This note combines the exact split-jet auxiliary theorem with the corrected induced-gravity coefficient. It states the strongest complete GR result that is presently justified: a local, generic-curvature, torsion-free Einstein infrared branch exists as a mathematically closed effective theory under an explicit spectral/Hessian assumption. The note also records why this is not yet an unconditional single- Θ derivation.

Let e be the physical tetrad, $\hat{\omega}(e)$ its Levi-Civita connection, and let S_{aux} be the split-jet multiplier action of `gap_10t_split_jet_auxiliary_completion.tex`. Under the specified one-loop assumptions of `gap_10d_induced_gravity_endgame.tex`, the infrared effective action through two derivatives is

$$\Gamma_{\text{IR}} = -\frac{1}{16\pi G_{\text{ind}}} \int d^4x \sqrt{g} (R - 2\Lambda_{\text{ren}}) + S_{\text{aux}} + S_{\text{matter}} + O(R^2/\Lambda^2, \nabla^4/\Lambda^2). \quad (1)$$

Here G_{ind} is the exact regulated expression

$$\frac{1}{G_{\text{ind}}} = \frac{N_B(1 - 6\xi)}{12\pi} \int_{\Lambda^{-2}}^{\infty} ds s^{-2} e^{-m^2 s} \vartheta_3(0, e^{-s/R_\psi^2}). \quad (2)$$

Theorem 1 (Conditional effective GR closure). *On a non-null split-jet patch, variation of Eq. (1) gives:*

1. *the exact split-jet representation $\hat{D}_\mu X = \sqrt{N_0} e_\mu$;*
2. *vanishing multiplier $\lambda = 0$, so the jet variables are nonpropagating and do not source the metric or physical spin connection;*
3. *the torsion-free Levi-Civita physical connection;*
4. *the Einstein equation*

$$G_{\mu\nu} + \Lambda_{\text{ren}} g_{\mu\nu} = 8\pi G_{\text{ind}} T_{\mu\nu} + O(\nabla^4/\Lambda^2). \quad (3)$$

The allowed local curvature is as general as in GR because the split-jet map is a right inverse for every tetrad.

Theorem 2 (GR recovery versus prediction of G). *Recovery of general relativity as an infrared effective theory requires a finite positive renormalized coefficient G_{R} in the Einstein equation; it does not require the low-energy theory to predict the numerical value of that dimensionful coefficient. Equation (1) therefore closes the conditional GR-recovery problem. Predicting G_{R} from the fundamental Θ measure is a strictly stronger UV-completion problem and remains open.*

Proof. The auxiliary Euler-Lagrange theorem gives the constraint and $\lambda = 0$. All auxiliary contributions to the tetrad and connection equations therefore vanish on shell. The remaining two-derivative variation is the standard variation of the Einstein-Hilbert functional. Universal local representability follows from the split-jet right inverse. $\square$

Strongest justified conclusion

UBT has a complete conditional effective GR branch: generic local GR kinematics, nonpropagating split-jet auxiliaries, and Einstein–Lambda infrared dynamics with an explicitly computable induced Newton coefficient.

It is not yet an unconditional single-field derivation: the current axioms do not derive the gauge-fixed Hessian, physical mode count, curvature coupling, UV cutoff identification, or constrained path-integral measure. Those are exactly the remaining first-principles inputs.